Ysbyty Gwynedd
Anaesthetics Department
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CATEGORY

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GUIDELINE FOR WARD MANAGEMENT FOLLOWING INTRATHECAL OPIOIDS.

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Key Points

- Intrathecal opioids may provide effective pain relief for up to 20-48 hours post procedure.
- For otherwise reasonably healthy patients an intrathecal dose of diamorphine up to 1mg may be used.
- Post operative monitoring following IT diamorphine follows that similar to patients in whom a PCA is used.
- If patients require more frequent monitoring or have received more than 1mg of diamorphine then ward based care is not appropriate.
- IT opioids provide highly effective and prolonged perioperative pain relief for thoracic, abdominal, orthopaedic and pelvic surgical procedures.
- The role of the epidural technique is decreasing and that of IT is increasing, particularly as part of multimodal analgesia strategy in ERAS protocols.
- Post operatively, patients can be nursed in regular surgical wards

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Introduction

The use of intrathecal (IT) opioids with or without local anaesthetics (LA) is a popular analgesic technique around the world for the management of post-operative pain (De Sousa & Chandron, 2014; Hindle, 2008). A 2008 UK survey showed that IT opioids were used routinely for intra and post-operative analgesia in almost 90% of the hospitals surveyed (Giovanelli et al, 2008). It is one of the easiest, most reliable and cost-effective methods for pain relief (Rawel, 2023).

Intrathecal opioids are highly effective in the management of post-operative pain. The technique is simple with a very low risk of technical failure or complications, and it does not require additional training or expensive equipment such as ultrasound machines and, therefore, is widely practised around the world. The high-quality pain relief is not associated

with sensory, motor or autonomic deficits. The use of IT is associated with prolonged analgesia lasting 20–48 h after a variety of surgical procedures. IT has a well-established role in thoracic, abdominal, spinal, urological and orthopaedic surgeries. It is considered the 'gold standard' analgesia technique for caesarean delivery which is generally performed under spinal anaesthesia. As the role of epidural technique in post-operative pain management continues to decrease, IT has emerged as the neuraxial technique of choice for pain management after a major surgery as a component of multimodal analgesia in Enhanced Recovery After Surgery (ERAS) protocols. The contraindications, complications and management of complications of intrathecal analgesia is similar to epidural analgesia. In general, the benefits of intrathecal morphine are limited to only slightly better pain relief for up to 24 hours after surgery and, while there can be reduction in the amount of systemic opioid required for up to 48 hours, the incidence of opioid-related adverse effects may not be decreased, in some patients they will increase (Schug et al., 2021).

Limitations of guideline

- This guideline applies to all Anaesthetists/ Registered Medical Staff who administer spinal opiates
- All nursing staff who are required to monitor patients after this has been administered
- This guidance does not relate to epidural care, or other indwelling catheters or the administration of the other drugs via the intrathecal route.
- The guideline does not advise on the actual insertion of spinal opioids within theatres. It is the responsibility of the anaesthetist to ensure the patient is appropriate for the use of this technique.
- This guideline is not suitable for intrathecal **MORPHINE** which has a higher risk of delayed respiratory depression.

Indications for spinal diamorphine

- Surgery where patients are likely to experience moderate to severe pain in order to reduce postoperative opioid requirements
- Where there is doubt that the patient could not manage PCA properly or where avoidance of PCA is desirable

Cautions

- Age> 80
- Respiratory disease
- Raised BMI
- Liver and Renal Impairment
- Patients already on opioid therapy.

Contraindications

- Patient refusal
- General contraindications to regional anaesthesia e.g., coagulopathy, localised infection
- Severe pulmonary disease such as severe COPD with history of type II respiratory failure,
- Patients who are unable to give informed consent
- Any patient that is known to be allergic to local anaesthetic.

Patient Care Considerations including management of side effects

The following guidelines for routine management of patients given spinal morphine/ diamorphine should apply:

- Must be given only by appropriately trained or supervised anaesthetist
- The dose and time of spinal diamorphine must be documented on the patient's anaesthetic chart
- All patients should be prescribed oxygen.
- A urinary catheter is recommended to avoid urine retention or incontinence embarrassment until mobility is restored.

Clinical Patient Group Criteria and Monitoring recommendations

Correct and careful monitoring of these patients is necessary to ensure they have adequate pain relief, but do not experience any side effects. These observations should be recorded with date and times on the dedicated observation chart.

Group 1:

- Patients < 80 years of age who receive IT diamorphine dose ≤ 1mg
- No major concern over other co-morbidity (anaesthetic assessment)

Plan:

- Suitable for ward-based care
- Patent I.V access

Observations:

- ***Every 15 minutes for 2 hours***
    - Blood Pressure
    - Heart Rate
    - Saturation
    - Respiratory Rate

  Then as before, in addition to:

    - Sedation
    - Nausea & vomiting
    - Pain Scores

- ***Hourly for 4 hours***
- ***4-hourly up to 24-hours postoperatively***

- Patient may also have:
    - PCA morphine/fentanyl for PCA
    - Oral opioids

Group 2

- 80 years
- IT Diamorphine dose 1mg <2mg
- Any patient with significant co-morbidity
- Patients already on strong opioids (MST, fentanyl patch etc) *Plan:*

As for Group 1:

- ***Every 15 minutes for 2 hours***
    - Blood Pressure
    - Heart Rate
    - Saturation
    - Respiratory Rate

Then as before, in addition to:

- Sedation
- Nausea & vomiting
- Pain Scores

Then:

- ***1 hourly for 12 hours***
- ***4-hourly up to 24-hours postoperatively***

THIS LEVEL OF CARE MAY PRECLUDE WARD-BASED CARE OF THESE PATIENTS.

Transfer from recovery to the ward

- Transfer to the ward will take place when the patient has been monitored closely for not less than half an hour and observations are stable.

- Patients should be prescribed oral opioids **OR** PCA by the Anaesthetist post-operatively depending on the type of surgery.

- Regular paracetamol and NSAID (if no contra-indication) should be prescribed and should be administered regularly.

- Two antiemetics should be prescribed – refer to local guidelines on management of post-operative nausea and vomiting.

- Patients may sit up slowly at any time following spinal anaesthesia/intrathecal analgesia. There is no time limit that they have to remain lying flat, unless they have a low BP, complain of headache, or for specific post-operative reasons related to the operation.

Management of Side Effects or Complications

Side effects are similar to those that occur with epidural opioids. The use of intrathecal opioids could result in developing common opioid side effects. These include pruritus, urinary retention, nausea and vomiting, with particular emphasis and concern about the possible development of delayed respiratory depression (Rathmell et al, 2005))

If Opioid Induced Ventilatory Impairment (OIVI) occurs following the administration of intrathecal morphine, the time-of-peak risk is about eight to ten hours after the injection,

although this can be much later. The time for peak risk of OIVI following intrathecal administration of highly lipid-soluble drugs is around 5-20 minutes.

Increasing patient age, high doses of intrathecal opioid, an opioid-naïve patient, and concurrent use of sedatives or systemic opioids are associated with an increased risk of OIVI.

Respiratory Depression

- If the respiratory rate is <8 per minute and the patient is not awake and not responding to voice use naloxone.
- Prepare naloxone 400 micrograms made up to 4 mL with sodium chloride 0.9%. Contact medical team to review patient.
- Administer naloxone 100 micrograms (i.e., 1 mL) every 1 minute until respiratory rate >10 and the patient are responding to voice.
- The effect of naloxone may wear off in less than 1 hour and further doses may be required.
- If respiratory depression develops, avoid further opioid doses, remove PCA handset from patient's reach. Seek advice from the pain team/on-call anaesthetist if pain remains a problem.

Sedation score 0-1

- Sedation is the earliest sign of respiratory depression.
- If the patient is not responding to voice, maintain airway if required, give oxygen using a facemask with reservoir bag at a flow rate of 15 litres/minute and call
- medical assistance.
- Prepare naloxone 400 micrograms made up to 4 mL with sodium chloride 0.9%. Administer 100 microgram (1ml) increments every 1 to 2 minutes up to a maximum dose of 400mcg every 1 minute until respiratory rate >10 and the patient is responding to voice
- Call Acute Pain Team (Bleep 115) or on-call Anaesthetist (063) if out of hours.

Nausea and vomiting

- Administer the prescribed antiemetics as per protocol for postoperative nausea and vomiting.

Pruritis

- Chlorphenamine orally 4-6mg 4-6 hourly maximum 24 mg/ 24 hours or 12 mg in elderly) should be offered IV 10 mg QDS maximum
- If severe itching is to chlorphenamine consider ondansetron 4mg stat IV (unlicensed use) followed by naloxone 20 microgram IV titrating up to 2 micrograms/kg IV if necessary and contact on-call anaesthetist (Bleep #063).

Urinary retention

- If the patient has not passed urine for 12 hours postoperatively and is complains of pelvic discomfort, contact medical staff.
- Urethral catheterization may be necessary if not already done.

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